

Consensus Algorithm for Fair Power Curtailment of PV Systems in LV Networks

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Abstract—Overvoltage problems are emerging as prominent issues all over Europe in low-voltage (LV) networks as they are hosting increasing share of distributed generation (DG) units. Active power curtailment with P-V droop control has been widely applied as a local solution to overvoltage issues. However, the conventional P-V droop control with constant droop parameters depend solely on local measurement and leads to unfair power curtailment among the connected end-users. In this paper, an innovative management scheme is proposed based on consensus algorithm for fair power curtailment. The algorithm focuses on the communication between adjacent nodes, while the control objectives are achieved by stepwise iterative calculations in the local controller. The algorithm has been tested through simulation analyses in three different LV networks. The simulation results indicate promising potential compared to conventional control methods.

Index Terms—Overvoltage, consensus algorithm, power curtailment, droop control.

I. INTRODUCTION

Global energy crisis and environmental issues have been promoting the roll-out of distributed renewable energy sources (DRES) [1], [2]. Distribution networks all over the world are hosting an increasing share of these distributed energy resources (DERs). Unlike conventional generation units, the generation of power of such DG units mainly takes place near the consumer, which can minimize the power losses in the distribution lines. However, RES-based DG units like solar PV are usually non-dispatchable and offer limited controllability during operation. Thus, capacity challenges in terms of voltage level violations and network congestions are becoming prevalent issues with RES-dominant networks. In radial LV networks, overvoltage problems appear along the feeder when local power generation exceeds the load and the generated power is injected into the network. The traditional approach of network reinforcement to address such issues may not be always feasible due to the scale of required investment. A number of alternative solutions have been studied, for instance with on-load tap-changer (OLTC) [3], reactive power compensation [4], active power curtailment [5], [6], demand response (DR) [7], and the application of electrical energy storage systems (EESSs) [8]. $P-V$ droop control of the PV inverters has been widely considered as a promising solution to address the overvoltage problems due to inherent simplicity of the mechanism. Such control schemes are usually

implemented locally at the inverter and do not require any external communication. However, using a fixed threshold and droop factor lead to unfair power curtailment, since the end-users towards the end of the feeder are curtailed more often due to the higher frequency of the problems [4], [5], [9]. A coordinated control for the $P-V$ droop is therefore required for a fairer allocation of curtailment.

Fair curtailment refers to an equal distribution of curtailed power among the end-users. There are two main methods to solve this kind of coordination problems: centralized control [10] and distributed control [11]. Centralized control relies on a centralized controller which processes the entire network information and sends the processed control command to each unit. The stability of such centralized controlled systems mainly depends on the applied communication technology. The reliability and robustness are relatively poor in case of a communication failure or changes in the network [10]. On the other hand, distributed control uses local communication between neighboring nodes to achieve the same control effect under centralized mode. The distributed control offers a flexible structure and strong robustness of the implemented system [11]. A distributed fair power sharing method is described in [9] based on network sensitivity calculation. Alternatively, distributed consensus algorithm can be used to solve the consistency problems of multiple end-users [12]. A similar distributed algorithm is proposed in [13] for reactive power redistribution and voltage regulation in traditional $Q-V$ droop control.

The core of the consistency theory is to achieve the consistent coordination of selected variables through communication between all adjacent nodes. The control objectives are achieved by stepwise iterative calculation in the local controllers [14].

This paper mainly focuses on the optimization of $P-V$ droop parameters by distributed consensus algorithm. The remainder of the paper is organized as follows: Section II discusses the background of the $P-V$ droop and distributed control and consensus algorithm; Section III explains the methodology proposed to solve the overvoltage problem in a fair way; Section IV describes the simulation setup; Section V presents the numerical results before summarizing with concluding remarks in Section VI.

II. CONSENSUS ALGORITHM FOR $P - V$ DROOP CONTROL

A. $P - V$ droop control

$P - V$ droop control has been widely applied for over-voltage mitigation in LV networks. Usually grid-tied PV inverters are controlled with maximum power point tracking (MPPT) algorithms to maximize the utilization of the solar irradiation [5], [15]. With $P - V$ droop control, power injected by the inverter depends on a function of the bus voltage (V) according to :

$$P_{inj} \approx \begin{cases} P_{MPPT} - m(V - V_{cri}), & V \geq V_{cri} \\ P_{MPPT}, & V < V_{cri} \end{cases} \quad (1)$$

where P_{MPPT} is the maximum power available in the PV panel for a given solar irradiance, m is a slope factor, V_{lim} is the limitation voltage above which the PV system should stop working and V_{cri} is the voltage (p.u) above which the power injected by the inverter is decreased with a droop factor. The slope factor, m can be obtained by:

$$m = \frac{P_{max}}{V_{lim} - V_{cri}} \quad (2)$$

B. Consensus algorithm for distributed control

Graph theory and matrix theory are the theoretical basis for studying distributed consistency problem. The connection relation between individuals in the system is generally described by the graph. Individuals are regarded as nodes in the graph, while the information communications between individuals are represented by the edges [16]. Usually, we use $G = (V, E)$ to express a graph, in which $V = \{1, 2, 3, \dots, n\}$ denotes the set of independent nodes, and $E \subset V \times V$ is used to represent the edges between nodes. The set of adjacent nodes of node i can be expressed by $N_i = \{j \in V | (j, i) \in E\}$ [11]. In a directed graph, the edges between nodes are oriented, i.e. $(j, i) \in E$ does not represent $(i, j) \in E$. Matrix theory describes the connection between nodes in a graph from a mathematical perspective. For directed graph G composed of n nodes, we usually use the adjacency matrix:

$$\mathbf{A} = [a_{ij}], \quad a_{ij} = \begin{cases} 1, & (j, i) \in E \\ 0, & \text{otherwise} \end{cases}$$

which represents the connection between nodes. If a certain weight is given to the edge of the graph, the 1 in the equation should be changed to the weight of the corresponding edge [12].

The feature of the consistency protocol is that the individuals in the system are consistent based on the ‘nearest neighbor rule’ (i.e., each individual only interacts with his neighbor and does not need to obtain global information) [17]. Thus, the consistency protocol is in fact, a distributed local protocol that can be expressed by:

$$|x_i - x_j| \leq \epsilon, \forall i, j \in V \quad (3)$$

For a distributed multi-agent system with n adjacent agents, the change of state of each agents depends on its current state and the current state of the adjacent agents, which can be expressed as:

$$\begin{cases} \dot{x}_i(t) = u_i(t) \\ u_i(t) = \sum_{j \in N_i} a_{ij}(x_j(t) - x_i(t)) \end{cases} \quad (4)$$

Defining the Laplacian matrix of the graph $\mathbf{L} = [l_{ij}]$, where:

$$l_{ij} = \begin{cases} -a_{ij}, & i \neq j \\ \sum_{j=1, j \neq i}^n a_{ij}, & i = j \end{cases} \quad (5)$$

we have the theorem that: If the matrix $\mathbf{L}$ remains constant in the control process of the multi-agent system, then all the state variable of the agents in the multi-agent system can reach a consensus gradually under the drive of the control law:

$$\begin{cases} \dot{X} = U \\ U = -LX \end{cases} \quad (6)$$

if and only if the graph G corresponding to $\mathbf{L}$ contains a directed spanning tree [12]. The directed spanning tree indicates that all the agents in the system can influence each other directly or indirectly.

The adjacency matrix $\mathbf{A}$ is a (0-1) matrix, so the elements in the Laplacian matrix are only related to the structure of the topology graph of the communication network. However, due to the particularity of the power system, the controllable capacity of each nodes/agents is different. If the same adjacency matrix element is used for the adjacent nodes, the convergence speed of the state variable will be limited [16]. Therefore, the corresponding weight z_{ij} can be given to adjust l_{ij} of the Laplacian matrix element, which can improve the convergence speed of the algorithm [17]. However, in order to reduce the fluctuation of iteration, the weight z_{ij} factor needs to meet the following requirements [11]:

$$\sum_{j=1}^n z_{ij} l_{ij} = 0 \quad (7)$$

Two typical variations of consensus algorithm exist, continuous consensus algorithm and discrete consensus algorithm. Considering the discrete control applied in most of the practical systems, we only introduce the first order discrete consensus algorithm here [18]:

$$x_i[k+1] = \sum_{j=1}^n d_{ij} x_j[k], \quad d_{ij} = \frac{z_{ij} |l_{ij}|}{\sum_{j=1}^n z_{ij} |l_{ij}|} \quad (8)$$

where $k = 1, 2, 3, \dots, n$ is the discrete time variable, $x_i[k]$ is the state value of node i after the k times iteration, d_{ij} are the components in the weighted Laplacian matrix which show the importance of the information collected from node i and j . If a precision requirement ϵ can be reached after iterations, the system is considered to be consistent.

In [19], a distributed adaptive droop control strategy based on discrete consensus was proposed to solve the unbalanced load-sharing problem of isolated DC micro-grid. The average voltage of the micro-grid was evaluated and an objective virtual resistance fulfilling the requirements was dynamically calculated by local controllers. The local controller would

communicate with its neighbors using voltage and current information within the synchronized clock, then the adaptive droop control based on variable virtual resistance was applied.

In [20], a distributed control strategy based on the discrete consensus algorithm was proposed. The upper layer strategy uses the consensus algorithm to optimize the output of PV systems to meet the total power demand and share the demand equally; the lower layer strategy applies voltage-based control method whose reference point is obtained by quadratic interpolation. The convergence speed of proposed strategy will be slow when the number of photovoltaic curves becomes large. To improve the convergence speed, Chebyshev polynomials filtering is used in the consensus algorithm.

III. METHODOLOGY

A. $P - V$ droop control as the local control

As discussed in the previous section, the $P - V$ droop control requires the information of the slope factor m , threshold voltage V_{cri} , the maximum power available from the PV system P_{MPPT} and the real time voltage at the given node, V_i . For the traditional droop control with constant droop parameters, the slope factor and threshold voltage are constant values. The maximum allowable voltage, V_{lim} is constant for all the nodes and is considered as 1.1 p.u. according to the EN50160 standard.

The droop control depends on the real time local voltage level. The control mechanism introduced above indicates that the local over-voltage problems can be tackled with the local droop control. A coordinated control algorithm can be designed on top of the local control to ensure the fairness of curtailment based on communication with neighboring systems.

B. Coordinated control algorithm

The utilization ratio of any PV system, α_i can be used to represent the percentage of injected active power after the curtailment:

$$\alpha_i = \frac{P_{inj}}{P_{MPPT}} \quad (9)$$

The goal of the coordinated operation is to ensure the fairness of curtailment among all the PV systems. In other words, the aim of the proposed control algorithm is to reach the same utilization ratio for each PV panel. For each node, we select the utilization ratio as the state variable and exchange the value of the variable with the adjacent nodes at each interval. The target utilization ratio can be obtained by first-order discrete consensus algorithm. This can be mathematically presented as:

$$\alpha_i[k+1] = \sum_{j=1}^n d_{ij} \alpha_j[k] \quad (10)$$

The method to choose proper information weights, d_{ij} has been studied by several previous studies [21]. In this paper, we choose a Metropolis method proposed in [22] to determine the values of d_{ij} , since the Metropolis weights have been well

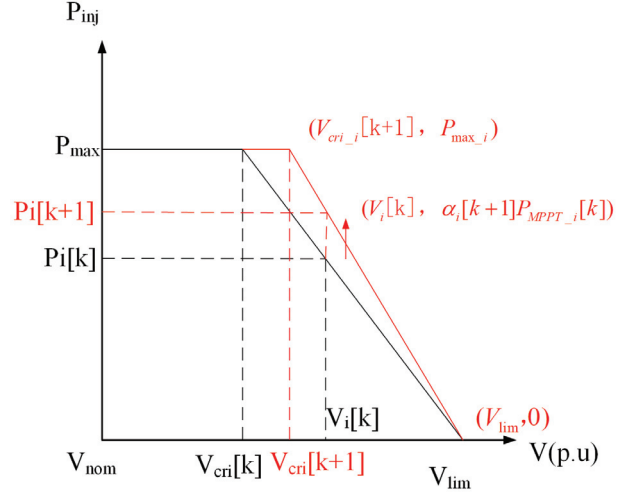


Fig. 1. Process of parameters updating for droop control.

studied and particularly suited for distributed implementation [23].

$$d_{ij} = \begin{cases} \frac{1}{\max(n_i, n_j)}, & j \in N_i \\ 1 - \sum_{j \in N_i} \frac{1}{\max(n_i, n_j)}, & i = j \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

where $\max(n_i, n_j)$ represents the larger number of neighbors owned by this node and its neighbors. This method satisfies the requirement of equations (6) and (7), which guarantees the reliability of the algorithm.

Figure 1 presents the process of calculating the parameters of the P-V droop for next time-step after the target utilization ratio for next period is determined. We have two points $(V_i[k], \alpha_i[k+1] * P_{MPPT-i}[k])$ and $(V_{lim}, 0)$. The parameters, the slope factor m and the threshold voltage V_{cri} can be therefore solved by:

$$P_{max-i} - m_i[k+1](V_{lim} - V_{cri-i}[k+1]) = 0 \quad (12)$$

$$P_{MPPT}[k] - m_i[k+1](V_i[k] - V_{cri-i}[k+1]) = \alpha_i[k+1]P_{MPPT} \quad (13)$$

$$\Rightarrow \begin{cases} m_i[k+1] = \frac{P_{max-i} - (1 - \alpha_i[k+1])P_{MPPT}[k]}{V_{lim} - V_i[k]} \\ V_{cri-i}[k+1] = V_{lim} - \frac{P_{max-i}}{m_i[k+1]} \end{cases} \quad (14)$$

The iterations stop when the nodes reach the local consistency. As discussed in Section II, local consistency is reached when the state values of a node and its adjacent nodes achieve become consistent, and is given by (3). We can also use $|\alpha_i[k+1] - \alpha_i[k]| < \epsilon$ as the criteria for the following reason:

$$\begin{aligned} \alpha_i[k+1] &= \sum_{j=1}^n d_{ij} \alpha_j[k] \\ &= (1 - \sum_{j \in N_i} d_{ij}) \alpha_i[k] + \sum_{j \in N_i} d_{ij} \alpha_j[k] \\ &= \alpha_i[k] + \sum_{j \in N_i} d_{ij} (\alpha_j[k] - \alpha_i[k]) \end{aligned}$$

Since ϵ only reflects the degree of accuracy, $|\alpha_i - \alpha_j| \leq \epsilon, \forall j \in N_i$ indicates $\sum_{j \in N_i} d_{ij}(\alpha_j[k] - \alpha_i[k])$ meets a precision requirement, that is $|\alpha_i[k+1] - \alpha_i[k]| < \epsilon$.

In this paper, V_{lim} is set as 1.1 p.u. The droop parameters will be updated according to the target utilization ratio, when local consistency protocol do not reach. Note that the utilization ratio cannot be larger than 1, it will be set to 1 when the target value becomes larger than 1.

IV. SIMULATION SETUP

The proposed methodology has been tested in a LV network (shown in Fig. 2) operated by French distribution system operator (DSO), SOREA. The nodes, where the PV systems and associated control algorithm are installed, have also been numbered and indicated in Fig. 2. For the remainder of the paper, notation for each node is composed by feeder and node number, e.g., first node of feeder 1 is denoted by 11. The network under study is supplied by a 20kV/400V, 20MVA transformer. The secondary voltage is raised to 420V in order to avoid under-voltage problems with heavy load. Since the feeders are composed of relatively short cable sections, lumped RL parameters can be used to model their characteristics. The loads have been modeled as three phase dynamic loads with constant power factor of 0.98. Average load profiles with 15 minutes time resolution have been used represent the active power consumption. The PV systems are simulated as negative dynamic loads with constant unity power factor. The network models have been developed in Simulink using SimPowerSystems Toolbox.

In order achieve fair power curtailment, we need to first determine how fairness needs to be treated. In this paper, two criteria for the fairness have been analyzed:

- 1) Total fairness, which means the communication happens among all the feeders, the same percentage of active power will be curtailed from all the nodes with the control, the closest nodes to the main bus of feeders can be considered as the adjacent nodes in this case;
- 2) Local fairness, which means the communication only takes place between the nodes at the same feeder i.e., the nodes at the same feeder will reach the same curtailed percentage of active power.

Simulations were performed for a typical summer day under three different cases: 1) without any control; 2) only with fixed P-V droop control; and 3) with coordinated control realized by consensus algorithm. For case 3, we will apply the consensus algorithm in two distinct ways: i) for the whole system, i.e., to reach the fairness for the whole network; and ii) for the feeders, i.e., to consider the fairness in each feeder rather than the whole LV network.

V. NUMERICAL RESULTS

In this section, the numerical results of the simulation case study will be presented. First, the impact of different levels of PV penetration on the network will be discussed. Next, the effects of the proposed algorithm will be elaborated in terms of its operation, voltage levels and fairness.

TABLE I
PERCENTAGE OF CURTAILED POWER IN TOTAL AND LOCAL CONSENSUS

Criteria	Feeder 1	Feeder 2	Feeder 3	Feeder 4	Feeder 5
Total	10.02	10.02	10.02	10.02	10.02
Local	0.84	3.51	4.22	2.25	27.91

A. Impact of different levels of PV penetration

The penetration level of PV has been expressed as the ratio of installed capacity and peak load of line section or feeder [24]. Fig. 3 depicts the maximum voltage in the network for different levels of PV penetration. For instance, with 100% penetration, the maximum voltage in the network reaches the upper allowable limit of 1.1 p.u.

B. Overvoltage mitigation

Fig. 4 shows the variation in voltage levels in all the nodes. Overvoltage problems are evident at the furthest two nodes of feeder 5 (56 and 57). Fig. 5 shows the voltage profiles and corresponding injected power at node 57, which has the most serious overvoltage problem in all three simulation cases. In both case 2 and case 3, the problem has been solved. However, case 2 requires the curtailment of more active power compared to case 3. Fig. 7 illustrates the percentage of curtailed energy for different cases. As can be seen, local droop control results in more curtailment of the active power for the nodes towards the end of the feeder. In contrast, the same percentage of active power has been curtailed from all the nodes in case 3.

C. Fairness evaluation

As illustrated in Fig. 7, the overall fairness in curtailment can be reached with total consensus and local consensus algorithms separately. Table I shows the total curtailment from each feeder with both criteria. It is obvious that more power will be curtailed with the total consensus algorithm except feeder 5. Since the power curtailed from the feeders can influence each other with the total consensus algorithm, the information from feeder 5 where overvoltage problem mainly happens, will be translated as more active power should be curtailed from other feeders.

D. Comparison of the communication protocols

Table II compares the percentage of curtailed power and the maximum voltage for the whole network for the three simulation cases in feeder 5. Less power is curtailed with the local consensus algorithm while a better overvoltage mitigation effect can be reached.

TABLE II
POWER CURTAILED PERCENTAGE FROM FEEDERS WITH TWO CRITERIA

	Local droop	Total consensus	Local consensus
Total energy curtailed (kWh)	582.27	359.75	317.92
PV generation (kWh)	3590.36	3590.36	3590.36
Total curtailed percentage (%)	16.21	10.02	8.85
Maximum voltage (p.u.)	1.072	1.094	1.089

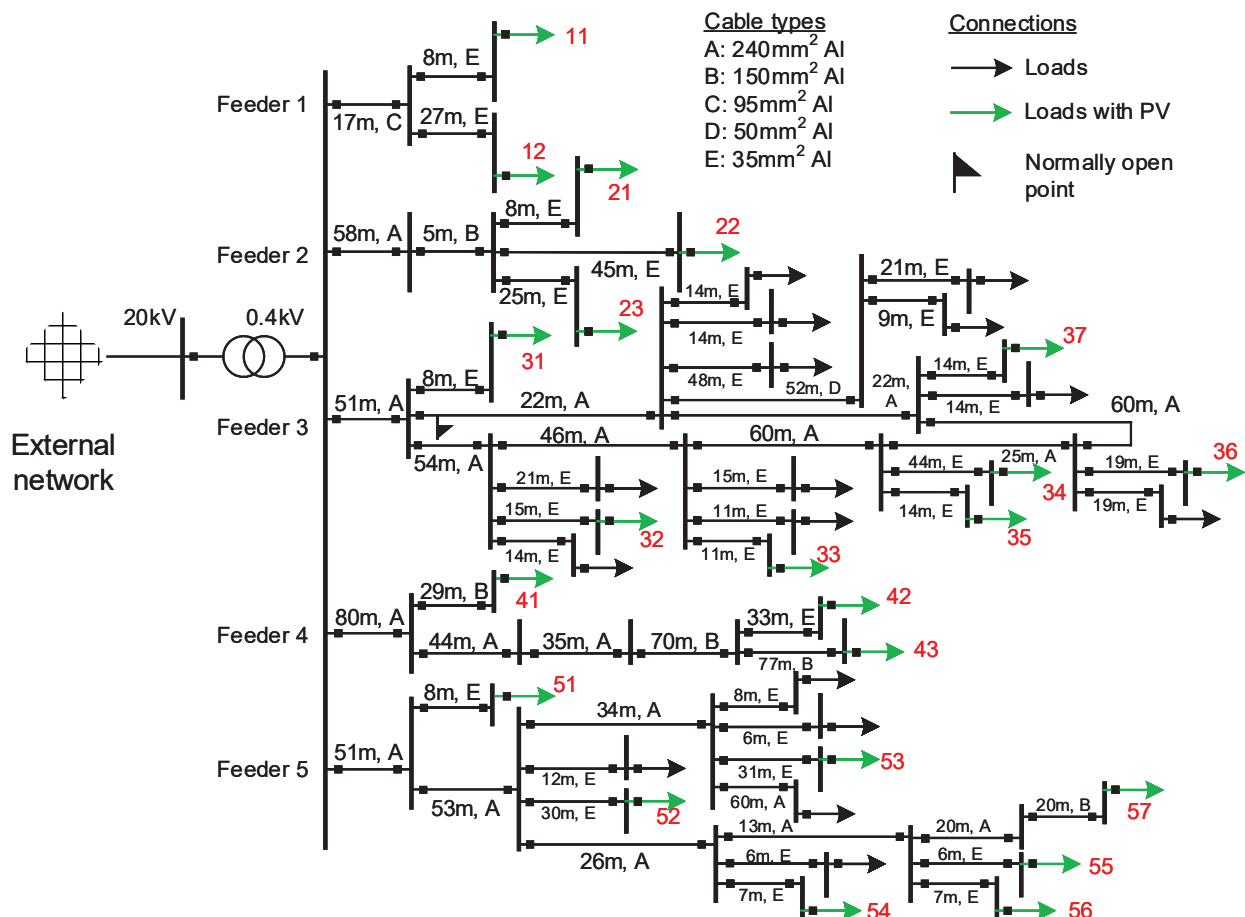


Fig. 2. Network model used in simulations. The numbers indicate the POCs with PV systems and associated control mechanism.

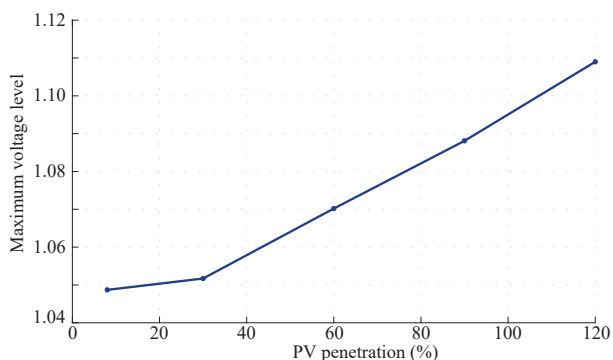


Fig. 3. Maximum voltage for different levels of PV penetration in the network.

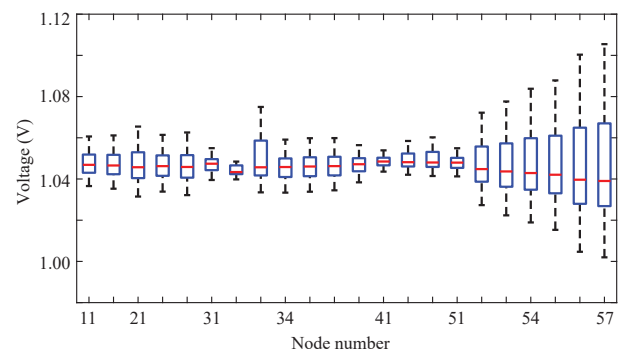


Fig. 4. Voltage variations in the nodes with PV systems.

VI. CONCLUSIONS

In this study, the consensus algorithm has been applied to solve the unfair power curtailment problem when $P - V$ droop is used as the local control for overvoltage mitigation in LV networks. The consensus algorithm has been applied considering two different criteria of fairness. The target fair power curtailment can be achieved by both of the proposed coordinated measures with consensus algorithm. The simu-

lation results show promising potentials for maximizing PV utilization in LV networks, i.e., the curtailed active power can be kept as minimum as possible while maintaining the voltage levels within allowable limits.

Future research on this topic will be directed to combine the $Q - V$ droop control with the existing $P - V$ droop control incorporating consensus algorithm. Integrated storage systems will also be investigated for a more flexible operation of the network as well as to maximize the revenue.

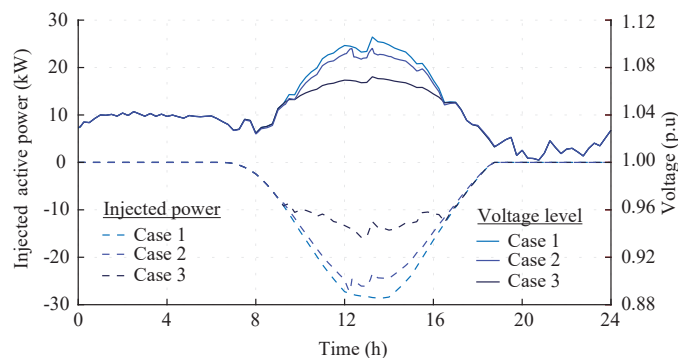


Fig. 5. Voltage variations and injected power at node 57 for three cases.

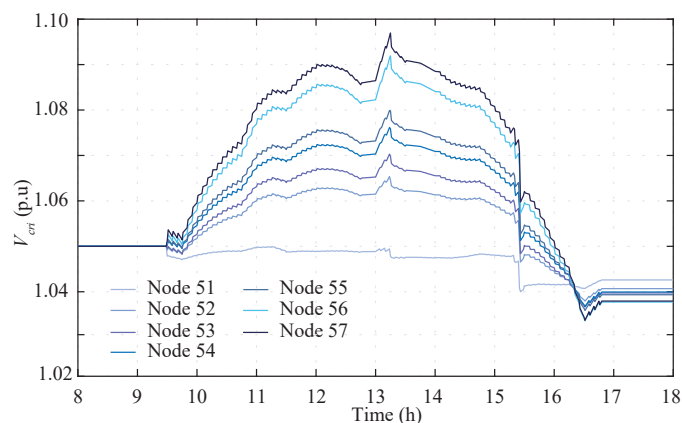


Fig. 6. Voltage threshold variation at all nodes of feeder 5.

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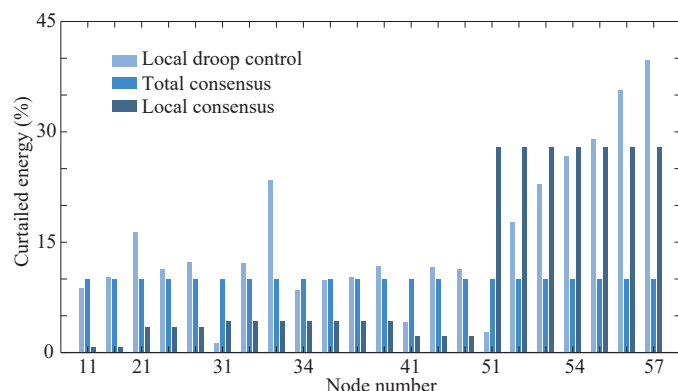


Fig. 7. Percentage of curtailed active power in three different cases.

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